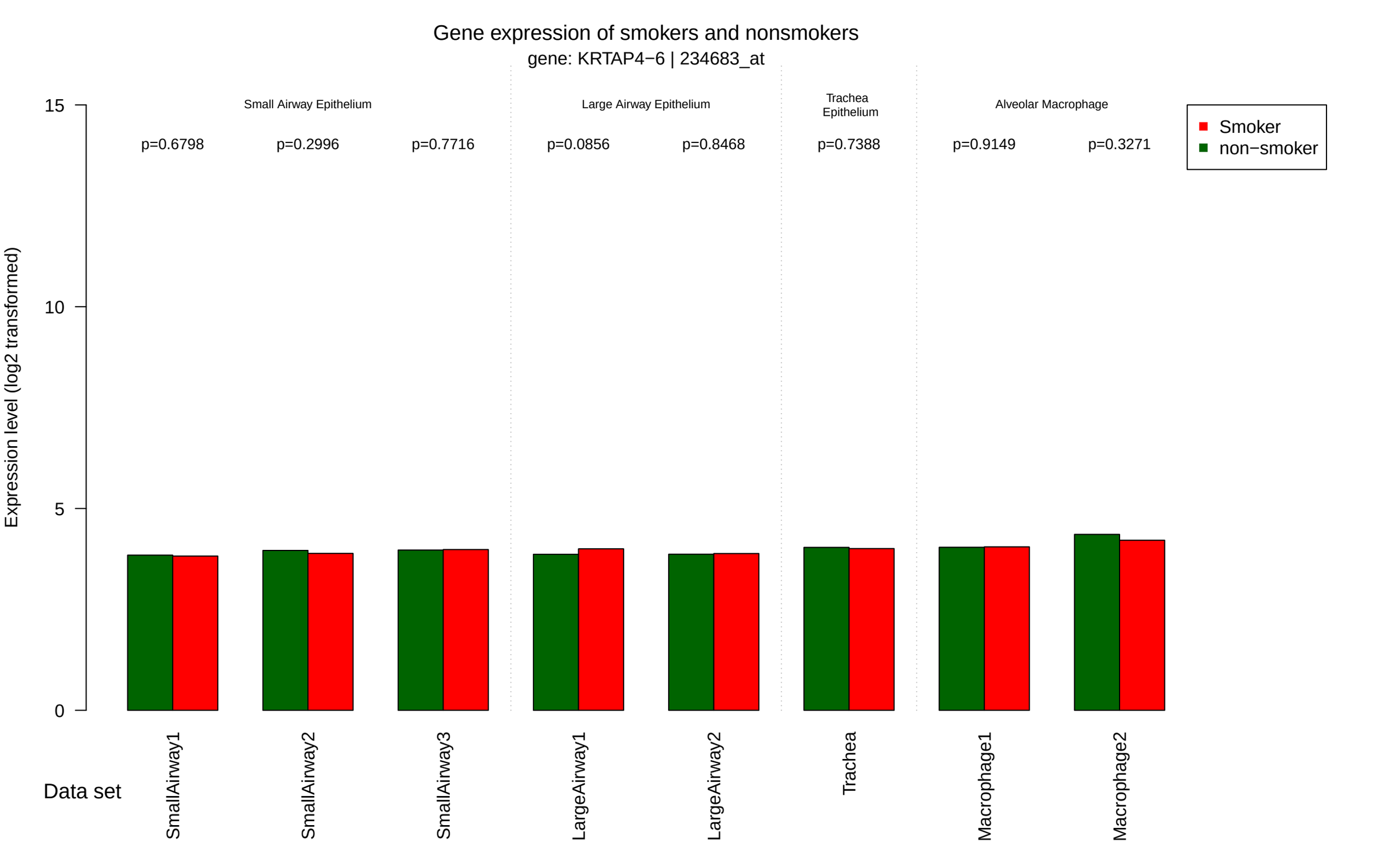
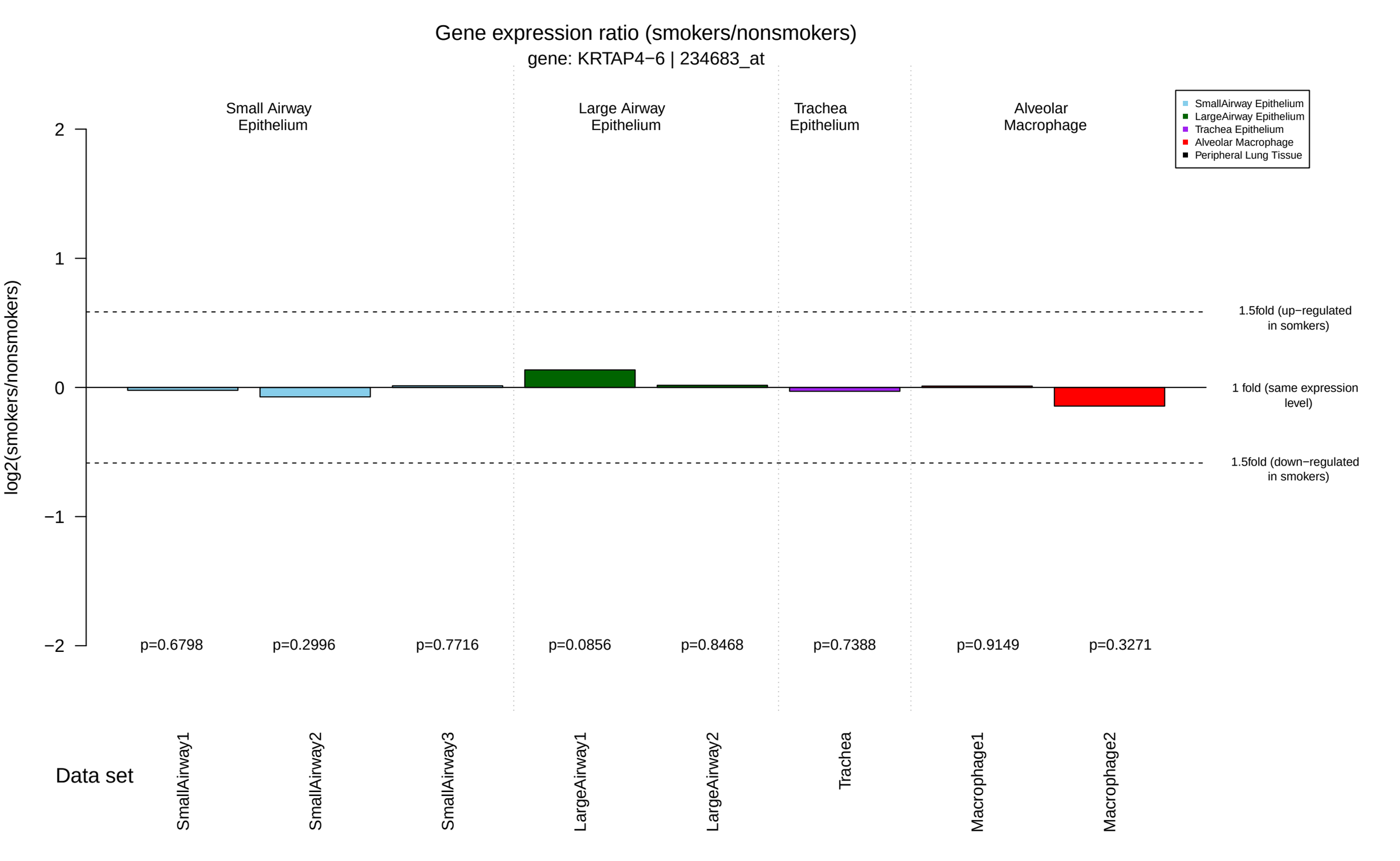
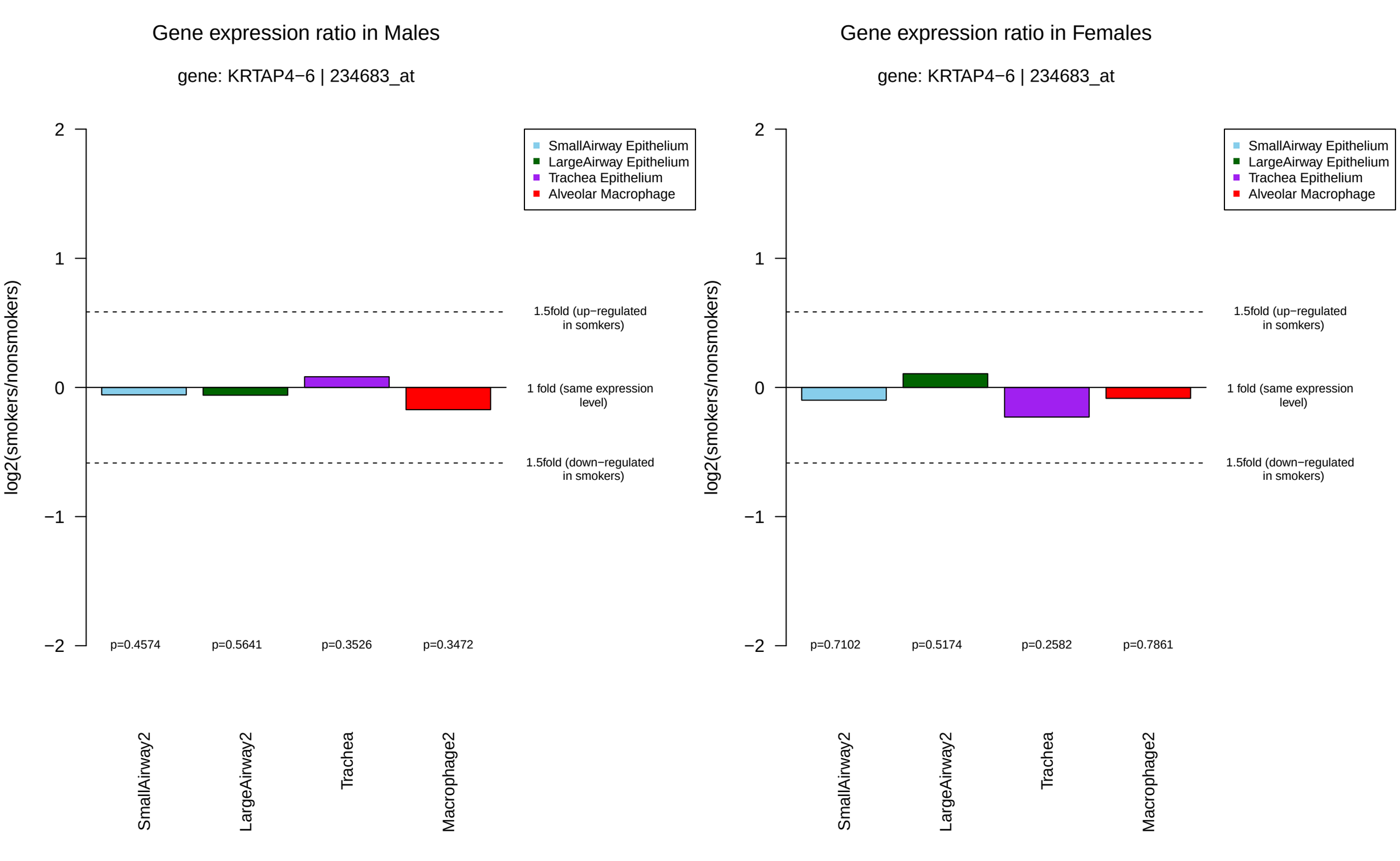
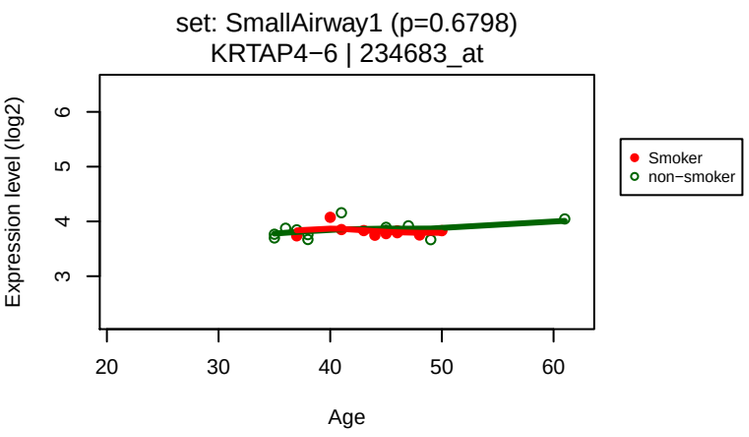
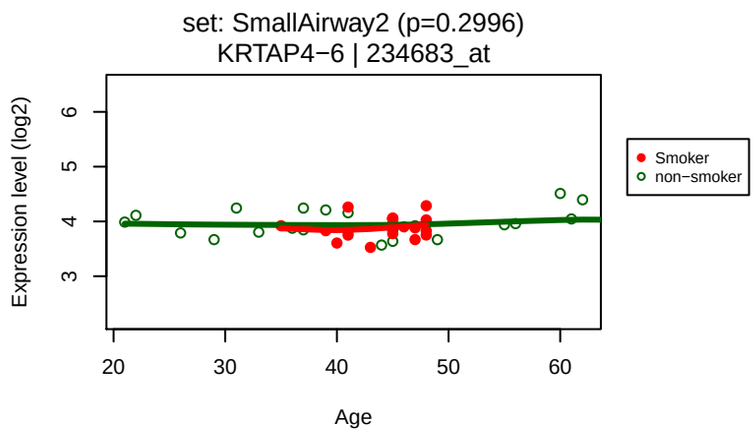
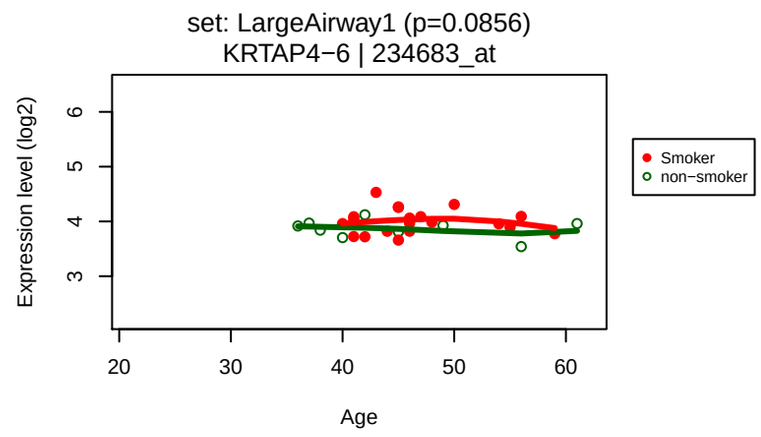

Gene expression of smokers and nonsmokers
gene: KRTAP4−6 | 234683_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6798
p=0.2996
p=0.7716
p=0.0856
p=0.8468
p=0.7388
p=0.9149
p=0.3271
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

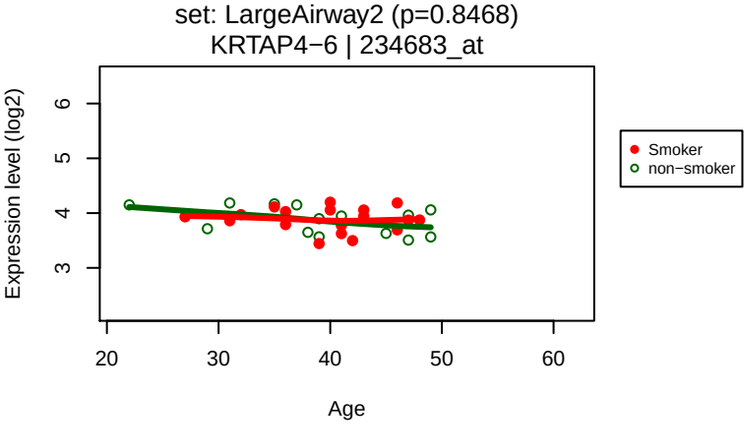

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP4−6 | 234683_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6798
p=0.2996
p=0.7716
p=0.0856
p=0.8468
p=0.7388
p=0.9149
p=0.3271
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

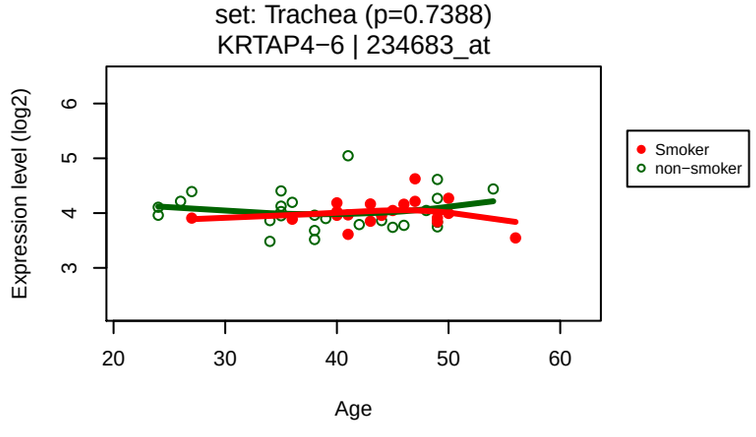

Gene expression ratio in Males
gene: KRTAP4−6 | 234683_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4574
p=0.5641
p=0.3526
p=0.3472
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP4−6 | 234683_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7102
p=0.5174
p=0.2582
p=0.7861
SmallAirway2
LargeAirway2
Trachea
Macrophage2

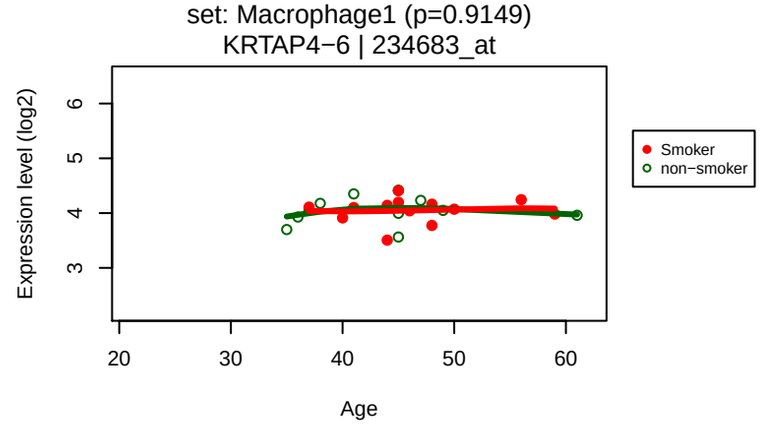

set: SmallAirway1 (p=0.6798)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker

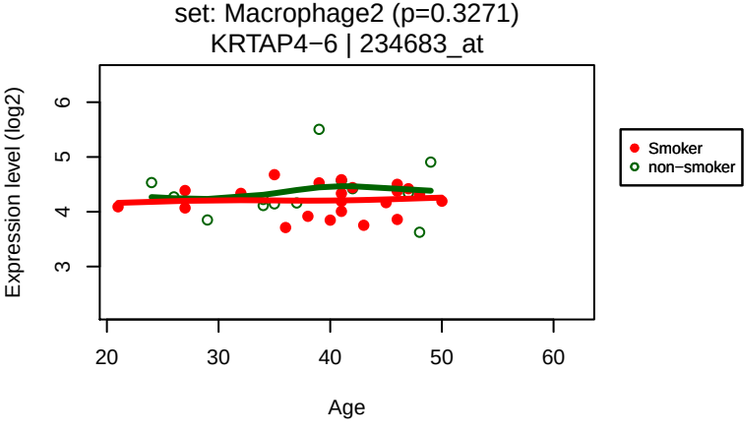

set: SmallAirway2 (p=0.2996)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker

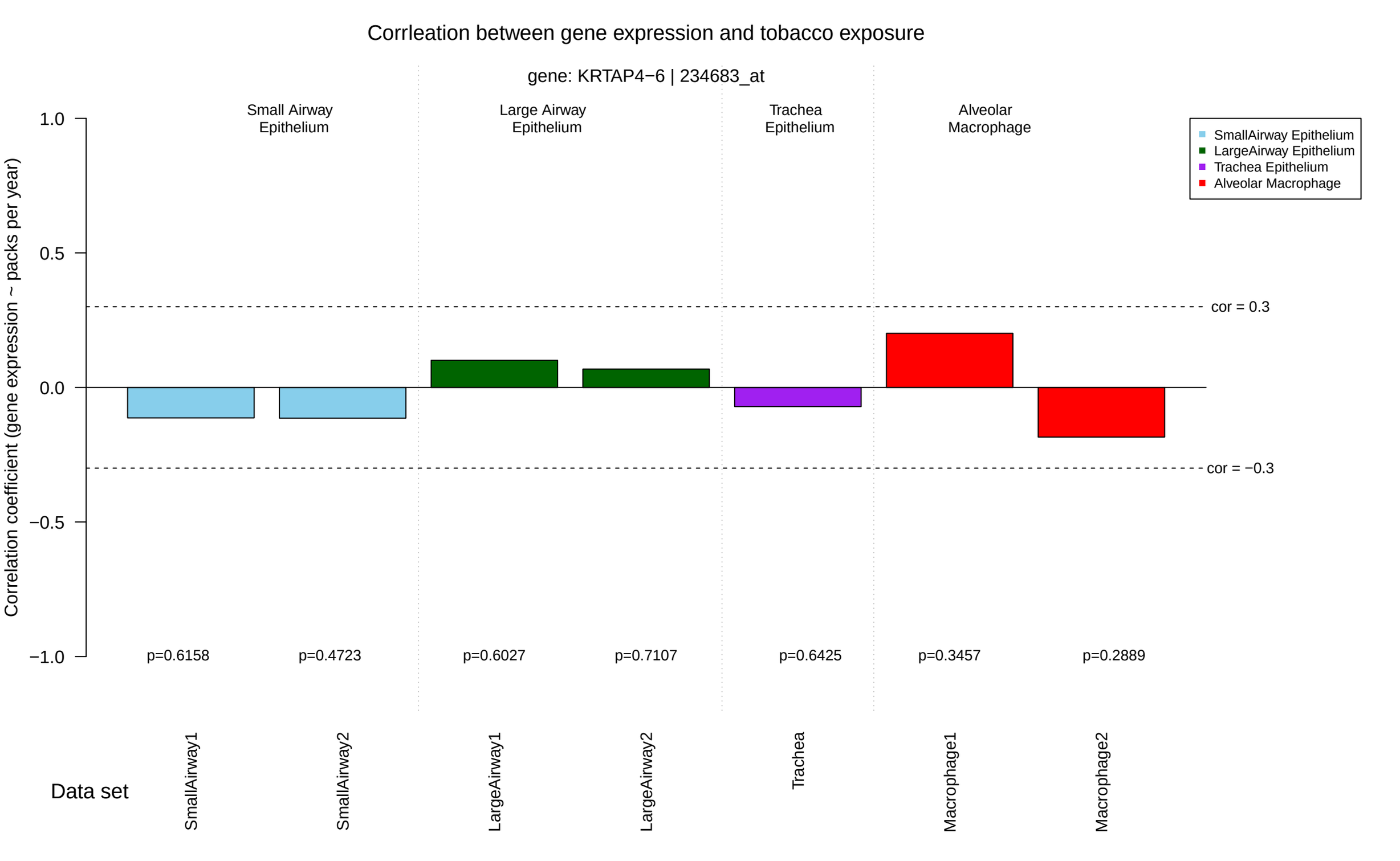

set: LargeAirway1 (p=0.0856)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8468)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7388)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9149)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3271)
KRTAP4−6 | 234683_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP4−6 | 234683_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6158
p=0.4723
p=0.6027
p=0.7107
p=0.6425
p=0.3457
p=0.2889
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2